

23CSE502

Software Testing

Module-1 Testing Principles and Axioms

Testing as a Process - Testing Axioms - Software Testing Principles - Origins and Cost of Defects - Defect Classes and Examples - Developer/Tester Support of Developing a Defect Repository - Defect Prevention Strategies.

Testing as a Process

The Role of Process in Software Quality

- The need for software products of high quality has pressured those in the profession to **identify and quantify quality factors such as usability, testability, maintainability, and reliability, and to identify engineering practices that support the production of quality products** having these favorable attributes.
- **The software development process** has been described as a series of **phases, procedures, and steps** that result in the production of a software product.
- Embedded within the software development process are several other processes including testing.

Testing as a Process

- The software development process has been described as a series of phases, procedures, and steps that result in the production of a software product, embedded within the software development process are several other processes including testing.
- Testing is described as a group of procedures carried out to evaluate some aspect of a piece of software.
- Testing can be described as a process used for revealing defects in software, and for establishing that the software has attained a specified degree of quality with respect to selected attributes.

Process in Engineering Domain

Process, in the software engineering domain, **is the set of methods, practices, standards, documents, activities, policies, and procedures** that software engineers use to develop and maintain a software system and its associated artifacts, such as project and test plans, design documents, code, and manuals.

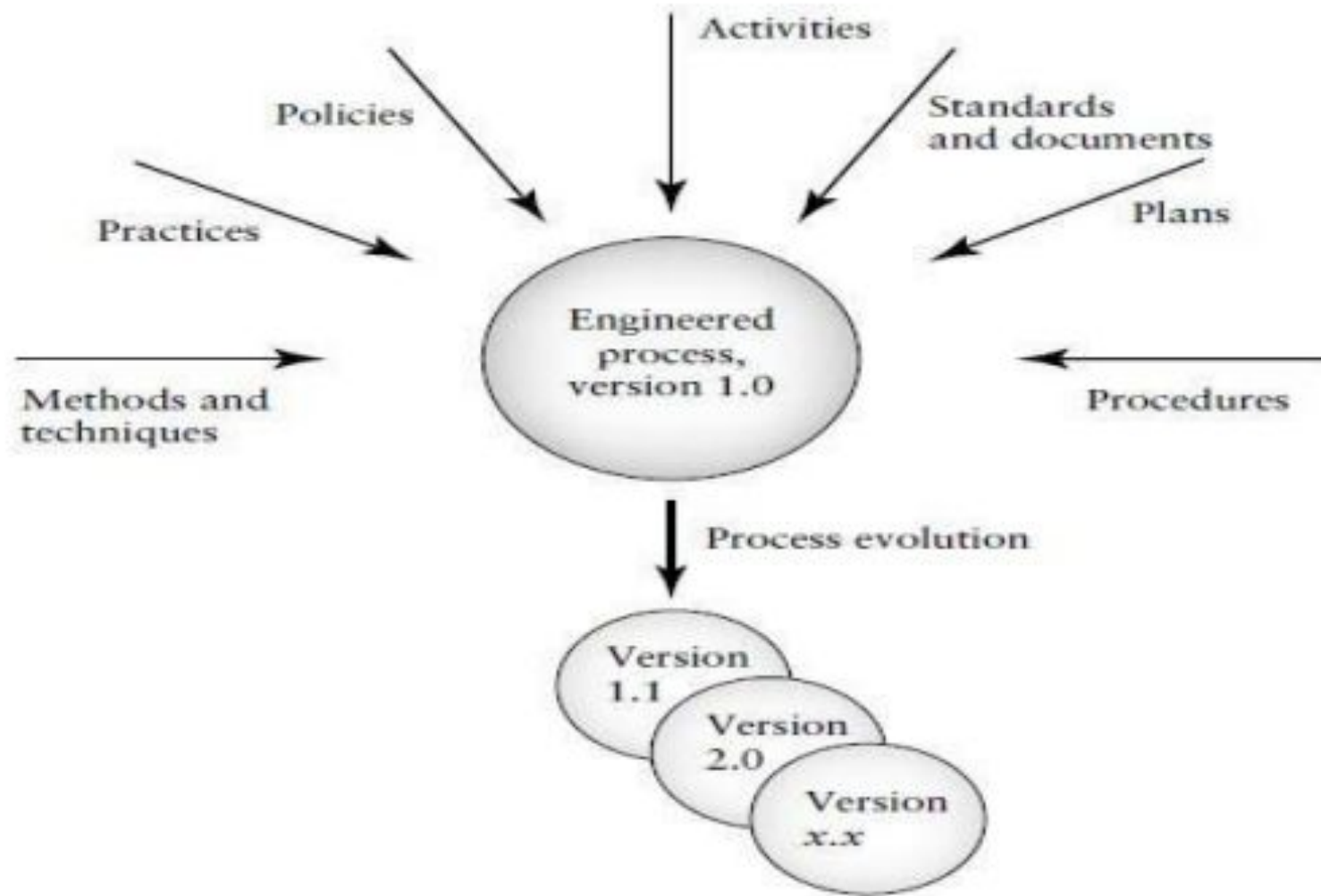


FIG. 1.2

Components of an engineered process.

■ Testing covers both validation and verification activities. Testing includes the following,

✓ Technical reviews

✓ Test planning

✓ Test tracking

✓ Test case design

✓ Unit test

✓ Integration test

✓ System test

✓ Acceptance test

✓ Usability test

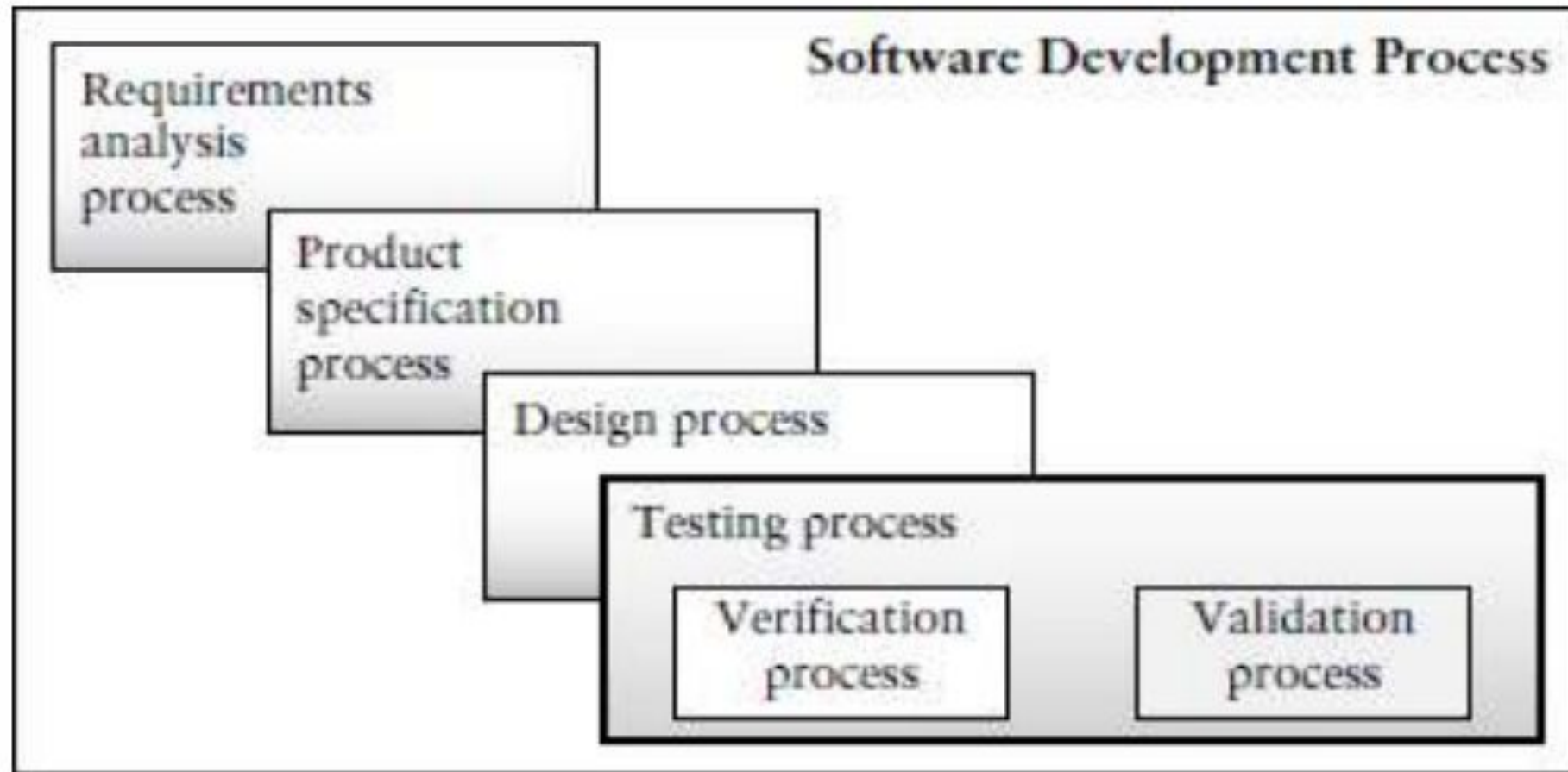
- Testing can also be described as a dual-purpose process.
- It reveals defects and evaluates quality attributes of the software such as:
 - Reliability

- Security
- Usability
- Correctness
- The debugging process begins after testing has been carried out and the tester has noted that the software is not behaving as specified.
- Debugging is the process of:
 - Locating the fault or defect
 - Repairing the code
 - Retesting the code
- Testing has economic, technical and managerial aspects.

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- Testing must be managed. Organizational policy for testing must be defined and documented.

Testing is related to two other processes.

- Validation is the process of **evaluating a software system** or component during, or at **the end of, the development cycle** in order to determine whether it satisfies specified requirements.
- Verification is the process of **evaluating a software system** or component to determine whether the products of a given development phase **satisfy the conditions imposed at the start of that phase.**



Example process embedded in the software development

Testing itself has been defined in several ways. Two definitions are shown below.

1. Testing is generally described as a group of procedures carried out to evaluate some aspect of a piece of software.
2. Testing can be described as a process used for revealing defects in software, and for establishing that the software has attained a specified degree of quality with respect to selected attributes.